What do three implementations teach us about scalability?

Lecture 9 · Compare

Evidence supports a conditional decision—not a universal winner.

Three builds create contrast, not a controlled ranking

Between A, B, and C, several things may change:

- model and harness;
- stack and architecture;
- student knowledge and practice;
- scope interpretation;
- hardware and deployment environment;
- orchestration and review process.

The shared specification and protocol make differences discussable, not perfectly attributable.

Build one evidence matrix across both systems

Compare each implementation across:

- required behavior and correctness;
- runtime pressure and resource use;
- architecture and failure recovery;
- setup, change, test, review, and operational friction;
- AI assistance and human correction;
- deployment and maintenance obligations;
- cost, sustainability, and known limits.

Missing evidence should remain visible.

Separate observation, inference, and proposal

Observation

What happened on the submitted revision under declared conditions?

Inference

What explanation is consistent with the evidence, and what alternatives remain?

Proposal

What should be changed or chosen next, given the goal and constraints?

Do not write an inference as though it were measured fact.

Name the threats to attribution

Before claiming that a tool or architecture caused a result, consider:

- practice effects between phases;
- unequal implementation scope;
- stack and dependency differences;
- hardware and service variation;
- measurement noise;
- missing failed attempts;
- review standards that changed over time.

A narrower claim is often the more useful engineering claim.

Defend a conditional decision

Use this structure:

> For **[goal and constraints]**, choose **[architecture and development workflow]** because **[strongest evidence]**. Reject **[alternative]** because **[trade-off or unsupported assumption]**. Revisit the decision when **[pressure or boundary]** changes.

The decision should name who owns its operational and correction obligations.

A strong defense answers uncomfortable questions

- Which result surprised you?
- Which claim has the weakest evidence?
- What did the scaling response make worse?
- Where did agents create review or integration load?
- Which difference cannot be attributed confidently?
- What would you measure next before committing more architecture?

Finish with the next boundary

The final package should contain:

- three reproducible implementations;
- comparable runtime and workflow evidence;
- one implemented and remeasured scaling response;
- a conditional architecture and workflow decision;
- rejected alternatives and remaining uncertainty;
- the next pressure, boundary, and measurement.

Scalability is not a finish line. It is a disciplined way to reason about change.